there would be a call $Explore(E''.p, \cdot, \cdot)$ with $E''$ a prefix of $E'_q$ and $p \in Sleep'$, and defining $w''$ by $E''.w'' = E'_q$, we would have $E'' \vdash p \diamond w''$ and $p \in WI_{[E'_q.q]}(w'_q.w_R.z')$, thus contradicting $WI_{[E'_q.q]}(w'_q.w_R.z') \cap Sleep' = \emptyset$. By the inductive hypothesis for $E'_q.q$ applied to $w'_q.w_R.z'$, the algorithm then explores some sequence $E'_q.q.z$ in $[E'_q.q.w'_q.w_R.z']_{\simeq}$. □

By the construction of $w_R$, we have $next_{[E'_q]}(q) \precsim_{E'_q.q.w'_q.w_R.z'} e_q$. From $E'_q.q.z \simeq E'_q.q.w'_q.w_R.z'$, it follows that the same race between $next_{[E]}(q)$ and $e_q$ will also occur in $E'_q.q.z$, i.e., we have $next_{[E'_q]}(q) \precsim_{E'_q.q.z} e_q$. Since the sequence $E'_q.q.z$ is actually explored by the algorithm, it will encounter the race $next_{[E'_q]}(q) \precsim_{E'_q.q.z} e_q$. When handling it,

- $E$ in the algorithm will correspond to $E'_q.q.z$ in this proof,
- $e$ in the algorithm will correspond to $next_{[E'_q]}(q)$ in this proof, and
- $e'$ in the algorithm will correspond to $e_q$ in this proof,
- $v = (notdep(next_{[E'_q]}(q), E'_q.q.z).\widehat{e_q})$ will be the sequence $v$ at line 5 in Algorithm 2.

Claim 7.9. $w'_q.w_R \sqsubseteq_{[E'_q]} v$.

Proof. Let

- $x$ be $notdep(next_{[E'_q]}(q), E'_q.q.z)$
- $x'$ be $notdep(next_{[E'_q]}(q), E'_q.q.w'_q.w_R.z')$. Note that $w'_q.w_q$ is a prefix of $x'$.

From $E'_q.q.z \simeq E'_q.q.w'_q.w_R.z'$ (Claim 7.8) and the definitions of $x$ and $x'$, it follows that $E'_q.x \simeq E'_q.x'$, and hence that $E'_q.x.\widehat{e_q} \simeq E'_q.x'.\widehat{e_q}$. Let $x''$ be obtained from $x'$ by adding $\widehat{e_q}$ just after the prefix $w'_q.w_q$ (i.e., in the same place that it has in $w'_q.w_R.z'$). Since $e_q$ "happens after" $next_{[E'_q]}(q)$ in $E'_q.q.w'_q.w_R.z'$, it follows that no events in $x'$ "happen after" $e_q$ in $E'_q.q.w'_q.w_R.z'$. Hence $E'_q.x'' \simeq E'_q.x'.\widehat{e_q}$. Since $w'_q.w_R$ is a prefix of $x''$ we have $w'_q.w_R \sqsubseteq_{[E'_q]} x''$, which by $E'_q.x'' \simeq E'_q.x'.\widehat{e_q}$ implies $w'_q.w_R \sqsubseteq_{[E'_q]} x'.\widehat{e_q}$, which by $E'_q.x.\widehat{e_q} \simeq E'_q.x'.\widehat{e_q}$ implies $w'_q.w_R \sqsubseteq_{[E'_q]} v$. □

Claim 7.10. $sleep(E'_q) \cap WI_{[E'_q]}(w'_q.w_R) = \emptyset$.

Proof. Assume that some process $p$ is in $WI_{[E'_q]}(w'_q.w_R)$. Then:

(1) If $p \in w'_q$, then it has no event happening before it, which implies that it cannot have been in $sleep(E'_q)$ since then it could not have been taken out of the sleep set. Thus $p \notin sleep(E'_q)$.
(2) Therefore $p \notin w'_q$. By $p \in WI_{[E'_q]}(w'_q.w_R)$ we have that $E'_q \vdash p \diamond w'_q$, which assuming $p \in sleep(E'_q)$, means that $p$ will still be in the sleep set after $w'_q$ and therefore $p \in final_sleep(E)$. Then:
    (a) If $p \in w_R$, then $p \in I_{[E]}(w_R)$ from which we get $p \in I_{[E]}(w)$ therefore $p \in WI_{[E]}(w)$. Since $p \in final_sleep(E)$ this contradicts Claim 7.5.
    (b) Therefore $p \notin w_R$ must hold. From $p \in WI_{[E'_q]}(w'_q.w_R)$ and $p \notin w'_q.w_R$ we have that $E'_q \vdash p \diamond w'_q.w_R$, which implies $E'_q.w'_q \vdash p \diamond w_R$, which is equivalent to $E \vdash p \diamond w_R$. But then $w_R = w_q.\widehat{e_q}$ is a prefix of $w_p$, implying that $w_p$ is strictly longer than $w_q$. This contradicts the fact that $q$ was chosen as the process in $final_sleep(E)$ with the longest prefix $w_q$ satisfying $E \vdash q \diamond w_q$.

Therefore, there can be no such $p \in WI_{[E'_q]}(w'_q.w_R)$ and Claim 7.10 is proven. □

From Claim 7.9, Claim 7.10, and Property 4.7(4) we get $sleep(E'_q) \cap WI_{[E'_q]}(v) = \emptyset$. Thus, the test at line 6 will succeed, and after performing line 7, the wakeup tree $wut(E'_q)$ will (by the specification of *insert*) contain a leaf $y$ such that $y \sim_{[E'_q]} v$. Since at this point $q \in wut(E'_q)$ and $q \notin WI_{[E'_q]}(v)$ we have, by definition of *insert*, that $E'_q \nvdash q \diamond y$. By Property 7.3(1) we then have $E'_q.y \in \mathcal{E}$.